Coursework 1: AI Storytelling Adventure Game

Undergraduate Individual Project

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1 Introduction

1.1 Project Overview

This project involves developing an AI-powered role-playing game (RPG) dungeon adventure that combines large language models (LLMs) for story generation with diffusion models for image creation. The goal is to create a fantasy adventure experience where players interact with an AI game master that generates both narrative text and corresponding visual scenes in real-time during gameplay.

1.2 Subject Area

The project spans multiple areas of computer science:

- **Artificial Intelligence & Machine Learning**: Integration of LLMs (Ollama/Llama3) for natural language generation and Stable Diffusion models (via ComfyUI) for image generation
- **Web Development**: Flask-based web application with real-time client-server communication
- **Human-Computer Interaction**: Interactive game interface with typing effects, audio narration, and dynamic content loading
- **Game Development**: Procedural content generation, game state management, and player interaction systems

1.3 Project Aim

The primary goal of this project is to develop a functional storytelling adventure game that integrates modern generative AI technologies for both text and image generation. The project aims to demonstrate how these technologies can be combined effectively to create an engaging interactive gaming experience. The system should:

- Generate coherent, context-aware narrative content using LLMs
- Produce high-quality fantasy-themed images that match the narrative scenes
- Provide an engaging, responsive user interface
- Handle real-time synchronization between text and image generation
- Maintain game state and player progression

1.4 Objectives

To achieve this goal, the following objectives have been established:

1. **AI Integration**: Successfully integrate Ollama (Llama3) for text generation and ComfyUI (Stable Diffusion) for image generation
2. **Web Application Development**: Build a Flask-based web server with session management and API endpoints

3. **Real-time Synchronization**: Implement mechanisms to synchronize text typing animation with image generation timing
4. **User Interface**: Design and implement an immersive, retro-styled game interface with typing effects, audio narration, and dynamic image loading
5. **Game Logic**: Develop a game engine capable of managing player state, choices, and narrative progression
6. **Performance Optimization**: Implement image caching and asynchronous processing to ensure responsive gameplay

2 Motivation

2.1 Why This Project Matters

The combination of generative AI and interactive entertainment is an area of growing interest and potential. This project explores several important aspects:

2.1.1 Advancing AI in Gaming

Traditional game development typically requires extensive manual content creation, which is time-consuming and costly. This project investigates whether AI-generated content can create unique, personalized experiences for each player. The potential benefits include:

- Reducing development costs and time
- Enabling infinite, procedurally generated content
- Creating personalized experiences for each player
- Demonstrating practical applications of modern AI technologies

2.1.2 Technical Integration Challenges

Combining multiple AI systems for text and image generation in real-time presents several technical challenges that need to be addressed:

- **Latency Management**: Balancing generation quality with response time
- **Context Preservation**: Maintaining narrative coherence across multiple interactions
- **Resource Management**: Efficiently handling GPU-intensive image generation
- **Synchronization**: Coordinating asynchronous processes (text typing, image generation, audio)

2.1.3 Educational Value

This project provides practical experience with:

- Modern AI/ML frameworks and APIs
- Full-stack web development
- Real-time system design
- Performance optimization techniques
- User experience design

2.2 Personal Motivation

The project was chosen because it combines several areas of interest:

- **AI/ML**: Exploring cutting-edge generative AI technologies
- **Game Development**: Creating interactive, engaging experiences
- **Web Technologies**: Building modern, responsive web applications
- **Creative Technology**: Combining technical skills with creative expression

2.3 Research Questions

This project aims to investigate the following questions:

1. Can LLMs effectively serve as game masters for interactive narrative experiences?
2. How can text and image generation be synchronized to create cohesive experiences?
3. What are the performance characteristics and limitations of real-time AI content generation?
4. How can user experience be optimized when dealing with variable generation times?

3 Literature Review

3.1 Large Language Models in Game Development

Large Language Models (LLMs) have shown remarkable capabilities in natural language understanding and generation. Recent models like GPT-3 [1], GPT-4 [6], and open-source alternatives like Llama [10] have demonstrated proficiency in creative writing, dialogue generation, and context-aware responses.

Application to Interactive Fiction: The use of LLMs for interactive fiction and text-based games has been explored in systems like AI Dungeon [11], which demonstrated that LLMs can generate coherent, context-aware narratives. However, these systems primarily focused on text-only experiences.

Local LLM Deployment: The emergence of local LLM deployment tools like Ollama [5] has made it feasible to run LLMs on consumer hardware, enabling privacy-preserving and cost-effective AI applications. This is particularly relevant for game development where cloud API costs can be prohibitive.

3.2 Diffusion Models for Image Generation

Stable Diffusion [7] represents a significant advancement in text-to-image generation, enabling high-quality image synthesis from textual prompts. The model's open-source nature and efficiency have made it widely adopted.

ComfyUI Framework: ComfyUI [2] provides a node-based interface for Stable Diffusion workflows, enabling complex image generation pipelines. It supports various models, control mechanisms, and customization options.

Real-time Generation Challenges: Image generation with diffusion models typically takes 5-30 seconds depending on hardware and model complexity. This presents challenges for real-time applications where user experience depends on responsiveness.

3.3 Procedural Content Generation in Games

Procedural Content Generation (PCG) has been a staple of game development, from early roguelikes to modern games like Minecraft and No Man's Sky. Togelius et al. [9] provide a comprehensive survey of PCG techniques.

AI-Driven PCG: Recent work has explored using AI for PCG. Summerville et al. [8] survey AI-based PCG methods, noting the potential of neural networks for content generation.

Narrative Generation: The combination of narrative and visual generation is less explored. Most PCG research focuses on either level generation or narrative generation separately, with limited work on integrated systems.

3.4 Web-Based Game Development

Modern web technologies enable sophisticated game experiences. Flask [3] provides a lightweight framework for Python-based web applications, suitable for game servers. WebSockets and AJAX enable real-time communication between client and server.

Session Management: Flask's session management allows for stateful game experiences, maintaining player progress and game state across interactions.

3.5 User Experience in Interactive Systems

The timing and synchronization of content delivery significantly impacts user experience. Nielsen [4] established that response times should be under 1 second for immediate feedback, but acceptable delays can be longer for complex operations.

Progressive Loading: Techniques like progressive image loading and typing animations can mask generation delays while maintaining user engagement.

3.6 Gaps in Current Research

While individual components (LLMs, diffusion models, web games) are well-researched, there is limited work on:

- Real-time integration of text and image generation in interactive systems
- Synchronization strategies for multi-modal AI content generation
- Performance optimization for consumer-grade hardware
- User experience design for variable-latency AI systems

This project attempts to address these gaps by developing a working system that demonstrates how these technologies can be integrated in practice.

4 First Steps

4.1 Project Setup and Planning

The initial phase of the project has focused on research, planning, and basic setup. The following work has been completed:

4.1.1 Technology Selection and Research

After reviewing the literature and available tools, the following technology stack has been selected:

- **Backend**: Python 3.10+ with Flask framework - chosen for its simplicity and suitability for web-based games
- **Text Generation**: Ollama with Llama3 model - selected for local deployment and privacy benefits
- **Image Generation**: ComfyUI with Stable Diffusion - chosen for its flexibility and open-source nature
- **Frontend**: HTML5, CSS3, JavaScript - standard web technologies for maximum compatibility
- **Audio**: Web Speech API - browser-based solution that doesn't require server-side processing

4.1.2 Project Structure Initialization

A basic project structure has been created with the following directory layout:

```
rpg_dungeon_ai/
          app.py                  # Main Flask application (planned)
          game/                   # Game logic modules
                player.py           # Player class (basic structure)
                engine.py           # Game engine (placeholder)
                ai.py               # AI integration (placeholder)
                world.py            # World state (placeholder)
          templates/              # HTML templates directory
          static/                 # Static assets directory
          generate_images.py      # Image generation module (planned)
          requirements.txt        # Python dependencies
```

4.1.3 Initial Development Work

The following preliminary work has been completed:

- Basic Flask application structure created with a simple route handler
- Player class skeleton implemented with basic health and inventory attributes
- Project dependencies identified and documented in `requirements.txt`
- Initial research into Ollama and ComfyUI APIs and integration methods
- Basic HTML template structure planned for the game interface

4.2 Current Progress Summary

At this early stage, the project has focused on foundational work:

- Project structure and directory layout established
- Technology stack selected and researched
- Basic Flask application skeleton created
- Player class structure defined with initial attributes
- Dependencies identified and documented
- Initial planning for AI integration approaches

The next phase will involve implementing the core functionality to bring the system to a working state.

4.3 Next Steps

4.3.1 Core Implementation (Weeks 6-8)

The following core functionality needs to be implemented to create a working system:

1. **Flask Web Server Development**:
   - Implement session management for player state and conversation history
   - Create RESTful API endpoints (`/`, `/make_choice`, `/get_image/<scene_id>`)
   - Set up template rendering with Jinja2
   - Develop JSON API for asynchronous scene updates
2. **AI Text Generation Integration**:
   - Integrate Ollama API with Llama3 model (`llama3:latest`)
   - Develop context-aware system prompts with MMO game master persona
   - Implement conversation history management (last 6 messages for context)
   - Create text cleaning utilities (markdown removal, option extraction)
   - Add support for both numbered choice selection and free-text input
3. **AI Image Generation Integration**:
   - Integrate ComfyUI API via HTTP requests
   - Implement asynchronous image generation using Python threading
   - Develop prompt enhancement system that adds fantasy MMO styling based on scene keywords
   - Implement image caching using MD5 hash-based filenames
   - Create dynamic image loading with client-side polling mechanism
   - Set up file management: images generated in `assets/` directory, moved to `static/images/` for serving

4. **User Interface Development**:
   - Design and implement retro-styled monitor interface with custom CSS
   - Create typing animation effect (45ms per character)
   - Implement dynamic choice buttons generated from AI output
   - Add free-text input option for custom player actions
   - Integrate text-to-speech narration using Web Speech API
   - Develop player status display
   - Add music toggle and audio controls
5. **Synchronization Implementation**:
   - Implement parallel text typing and image generation (both start simultaneously)
   - Configure AI prompt to generate 600-700 character descriptions to match image generation time
   - Develop image polling mechanism (checks every 200ms initially, then 100ms after text completes)
   - Create smooth transitions between scenes with image fade-in effects

4.3.2 Technical Implementation Details

The following technical challenges will need to be addressed:

- **Asynchronous Processing**: Implement background threading for image generation to avoid blocking the Flask web server
- **Image Caching**: Develop MD5 hash-based filename system to cache generated images and avoid regeneration
- **Prompt Engineering**: Create intelligent prompt enhancement that analyzes scene text for keywords (town, dungeon, boss, etc.) and adds appropriate fantasy MMO styling
- **Timing Optimization**: Adjust AI prompts to generate longer text (600-700 characters) so typing animation (27-31 seconds) matches image generation time (around 28 seconds)
- **ComfyUI Integration**: Develop custom workflow JSON configuration for Stable Diffusion with configurable parameters
- **Error Handling**: Implement status checking before generation attempts and graceful fallbacks when services are unavailable

4.3.3 Expected System Workflow

Once implemented, the system workflow will be:

1. **Initialization**: Player session created, system prompt configured, initial scene and image generated
2. **Scene Display**: Text typing animation begins while image generates in parallel
3. **User Interaction**: Player selects options or types free text, submitted via AJAX
4. **Response Generation**: User choice added to history, sent to Ollama, new scene generated, image queued asynchronously
5. **Content Delivery**: New text and options returned as JSON, client updates scene, image displayed when ready
6. **State Management**: Player status and conversation history maintained in session

4.3.4 Anticipated Challenges

Several challenges are expected during implementation:

- **Image Generation Latency**: Image generation takes around 28 seconds, which may cause timing issues with text display. Solution will involve adjusting AI prompts to generate longer text descriptions.
- **Real-time Synchronization**: Coordinating text typing and image loading will require careful implementation of parallel processing with polling mechanisms.
- **Service Integration**: Ensuring reliable communication between Flask, Ollama, and ComfyUI services will require robust error handling.

4.3.5 Testing and Validation Plan

Once core functionality is implemented, the following testing will be conducted:

- Functional testing of Flask server, session management, and API endpoints
- Validation of Ollama integration for coherent narrative generation
- Confirmation of ComfyUI integration for appropriate image generation
- Testing of user interactions and error handling
- Performance testing to measure generation times and optimize synchronization

4.3.6 Future Enhancements (Weeks 9-12)

After core functionality is complete, potential enhancements include:

- Enhanced game mechanics (combat system, quest system, NPC interactions)
- Improved AI integration (fine-tuned prompts, context-aware image generation)
- User experience enhancements (save/load functionality, achievement system)
- Performance optimizations (image pre-generation, request queuing)
- Advanced features (multiplayer support, character customization, procedural generation)

4.4 Project Timeline

- **Weeks 1-5**: Project setup, research, planning, first coursework submission ✓
- **Weeks 6-8**: Core implementation (Flask server, AI integration, UI development, synchronization)
- **Weeks 9-10**: Testing, debugging, optimization, enhanced features
- **Weeks 11-12**: Advanced features, evaluation, documentation
- **Week 13**: Final submission preparation

5 Ethics Forms

5.1 Ethics Screening

This project has been reviewed and determined to be **LOW RISK** for the following reasons:

- **No Human Participants**: The project does not involve direct data collection from human participants
- **No Sensitive Data**: No personal, medical, or sensitive information is collected or processed
- **Public Data Only**: Uses publicly available AI models and open-source technologies
- **Local Deployment**: System runs locally; no external data transmission required
- **No User Tracking**: No analytics, tracking, or user behavior monitoring

5.2 Ethics Screening Form

The Research Ethics Screening Form (Version 8, December 2025) has been completed and submitted separately to the supervisor for review and approval. The signed form is included as a separate document with this coursework submission.
Supervisor: Daniele Dell'Erba
Date Submitted: 4th February 2026

6 References

References

[1] Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., ... & Amodei, D. (2020). Language models are few-shot learners. *Advances in neural information processing systems*, 33, 1877-1901.

[2] ComfyUI. (2024). *ComfyUI - A powerful and modular stable diffusion GUI and backend*. GitHub. `https://github.com/comfyanonymous/ComfyUI`

[3] Grinberg, M. (2018). *Flask Web Development: Developing Web Applications with Python*. O'Reilly Media.

[4] Nielsen, J. (1994). *Usability Engineering*. Morgan Kaufmann.

[5] Ollama. (2024). *Ollama - Run large language models locally*. `https://ollama.ai/`

[6] OpenAI. (2023). GPT-4 Technical Report. *arXiv preprint arXiv:2303.08774*.

[7] Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10684-10695.

[8] Summerville, A., Snodgrass, S., Guzdial, M., Holmgård, C., Hoover, A. K., Isaksen, A., ... & Togelius, J. (2018). Procedural content generation via machine learning (PCGML). *IEEE Transactions on Games*, 10(3), 257-270.

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[10] Touvron, H., Lavril, T., Izacard, G., Martinet, X., Lachaux, M. A., Lacroix, T., ... & Lample, G. (2023). LLaMA: Open and efficient foundation language models. *arXiv preprint arXiv:2303.08774*.

[11] Walton, N. (2019). *AI Dungeon*. Latitude Games. `https://www.aidungeon.com/`

Appendices

Appendix A: System Requirements

- **Operating System**: Windows 10/11
- **Python**: 3.10 or higher
- **GPU**: NVIDIA GPU with CUDA support (RTX 5090 recommended for optimal performance)
- **RAM**: 16GB minimum, 32GB recommended
- **Storage**: 50GB+ for models and dependencies
- **Network**: Internet connection for initial setup (local operation after setup)

Appendix B: Code Structure

Main Application Files

app.py - Main Flask application containing:

- Flask route handlers (`/`, `/make_choice`, `/get_image`)
- Session management for player state and conversation history
- Integration with Ollama for text generation
- Image generation coordination via threading
- Text processing utilities (markdown removal, option extraction)

generate_images.py - ComfyUI integration module:

- HTTP API communication with ComfyUI server
- Workflow JSON configuration for Stable Diffusion
- Image file management and caching
- Status checking and error handling

Game Logic Modules

game/player.py - Player class implementation:

- Health and inventory management
- Status tracking methods
- Item addition/removal functionality

game/engine.py, game/ai.py, game/world.py - Placeholder modules for future expansion:

- Currently empty, reserved for game engine logic
- Planned for combat system, world state management, and advanced AI features

Frontend Files

templates/game.html - Main game interface:

- HTML structure with embedded JavaScript
- Typing animation implementation
- Image loading and polling mechanism
- Text-to-speech integration using Web Speech API
- Dynamic choice button generation

static/styles.css - Retro-styled CSS for the game interface
static/images/ - Directory for cached generated images

Configuration Files

requirements.txt - Python package dependencies including Flask, Ollama, and image processing libraries
launch_rpg_dungeon.bat - Windows batch script for automated service startup

Word Count: Approximately 2,500-3,000 words

Last Updated: 4th February 2026

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Do not amend before use

Research Ethics Screening Form for Students

(For students on taught programmes only, NOT for PGRs - MRes/MPhil/PhD)

Middlesex University is concerned with protecting the rights, health, safety, dignity, and privacy of its research participants. It is also concerned with protecting the health, safety, rights, and academic freedom of its students and with safeguarding its own reputation for conducting high quality, ethical research.

*This Research Ethics Screening Form will enable taught students to self-assess and determine whether their research requires ethical review and approval via the online **Ethics Monitor** before commencing their study. Supervisors must approve this form after consultation with students.*

Student Name:	Dumitru Nirca	Email:	DN408@live.mdx.ac.uk
Research project title:	Interactive Storytelling Game Leveraging Generative AI		
Programme of study/module:	BSc Computer Science		
Supervisor Name:	Daniele Dell'Erba	Email:	D.DellErba@.mdx.ac.uk

Please answer whether your research/study involves any of the following given below:		
1. [H]ANIMALS or animal parts.	☐ Yes	☑ No
2. [M]CELL LINES (established and commercially available cells - biological research).	☐ Yes	☑ No
3. [H]CELL CULTURE (Primary: from animal/human cells- biological research).	☐ Yes	☑ No
4. [H]CLINICAL Audits or Assessments (e.g. in medical settings).	☐ ~~Yes~~	☑ No
5. [X]CONFLICT of INTEREST or lack of IMPARTIALITY. If unsure see "Code of Practice for Research" (Sec 3.5) at: https://unihub.mdx.ac.uk/study/spotlights/types/research-at-middlesex/research-ethics	☐ Yes	☑ No
6. [X]DATA to be used that is not freely available (e.g. secondary data needing permission for access or use).	☐ Yes	☑ No
7. [X]DAMAGE (e.g., to precious artefacts or to the environment) or present a significant risk to society).	☐ Yes	☑ No
8. [X]EXTERNAL ORGANISATION – research carried out within an external organisation or your research is commissioned by a government (or government body).	☐ Yes	☑ No
9. [M]FIELDWORK (e.g., biological research, ethnography studies).	☐ Yes	☑ No
10.[H]GENETICALLTY MODIFIED ORGANISMS (GMOs) (biological research).	☐ Yes	☑ No
11. [H]GENE THERAPY including DNA sequenced data (biological research).	☐ Yes	☑ No
12. [M]HUMAN PARTICIPANTS – ANONYMOUS Questionnaires (participants not identified or identifiable).	☐ Yes	☑ No
13. [X]HUMAN PARTICIPANTS – IDENTIFIABLE (participants are identified or can be identified): survey questionnaire/ INTERVIEWS / focus groups / experiments / observation studies/ evaluation studies.	☐ Yes	☑ No

14. [H]HUMAN TISSUE (e.g., human relevant material, e.g., blood, saliva, urine, breast milk, faecal material).	☐ Yes	☑ No
15.[H]ILLEGAL/HARMFUL activities research (e.g., development of technology intended to be used in an illegal/harmful context or to breach security systems, searching the internet for information on highly sensitive topics such as child and extreme pornography, terrorism, use of the DARK WEB, research harmful to national security).	☐ Yes	☑ No
16. [X]PERMISSION is required to access premises or research participants.	☐ Yes	☑ No
17. [X]PERSONAL DATA PROCESSING (Any activity with data that can directly or indirectly identify a living person). For example, data gathered from interviews, databases, digital devices such as mobile phones, social media or internet platforms or apps with or without individuals'/owners' knowledge or consent, and/or could lead to individuals/owners being IDENTIFIED or SPECIAL CATEGORY DATA (GDPR[1]) or CRIMINAL OFFENCE DATA. [1]Special category data (GDPR- Art.9): "personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation".	☐ Yes	☑ No
18.[X]PUBLIC WORKS DOCTORATES: Evidence of permission is required for use of works/artifacts (that are protected by Intellectual Property (IP) Rights, e.g. copyright, design right) in a doctoral critical commentary when the IP in the work/artifacts jointly prepared/produced or is owned by another body	☐ Yes	☑ No
19. [H]RISK OF PHYSICAL OR PSYCHOLOGICAL HARM (e.g., TRAVEL to dangerous places in your own country or in a foreign country (see https://www.gov.uk/foreign-travel-advice), research with NGOs/humanitarian groups in conflict/dangerous zones, development of technology/agent/chemical that may be harmful to others, any other foreseeable dangerous risks).	☐ Yes	☑ No
20.[H]SECURITY CLEARANCE – required for research.	☐ Yes	☑ No
21.[H]SENSITIVE TOPICS (e.g., anything deeply personal and distressing, taboo, intrusive, stigmatising, sexual in nature, potentially dangerous, etc).	☐ Yes	☑ No

M – Minimal Risk; X – More than Minimal Risk. H – High Risk

If you have answered 'Yes' to ANY of the items in the table, your application REQUIRES ethical review and approval using the online **Ethics Monitor BEFORE commencing your research**. Please apply for ethical review and approval using the **Ethics Monitor**. Further information such as links to access **Ethics Monitor** & training videos is available at MyLearning(*Middlesex Online Research Ethics) area*: https://mdx.mrooms.net/course/view.php?id=12277 (Login required).

If you have answered 'No' to ALL of the items in the table, your application is Low Risk and you may NOT require ethical review and approval using **Ethics Monitor** before commencing your research. Your research supervisor will confirm this below.

Student Signature: Dumitru Nirca Date: 19/02/2026

To be completed by the supervisor:

Based on the details provided in the self-assessment form, I confirm that:	Insert Y or N
The study is Low Risk and *does not require* ethical review & approval using **Ethics Monitor**	■
The study *requires* ethical review and approval using **Ethics Monitor**	

Supervisor Signature: Daniele Dell'Erba Date: Feb 20, 2026

Daniele Dell'Erba (Feb 20, 2026 10:47:06 GMT)

Research Ethics Screening Form-taught students-Ver8-Dec 2025 MDX

Final Audit Report 2026-02-20

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